

Chapter 1: Integers

Drill practice and mixed revision of all 10 lessons · 96 questions

NSW syllabus outcome

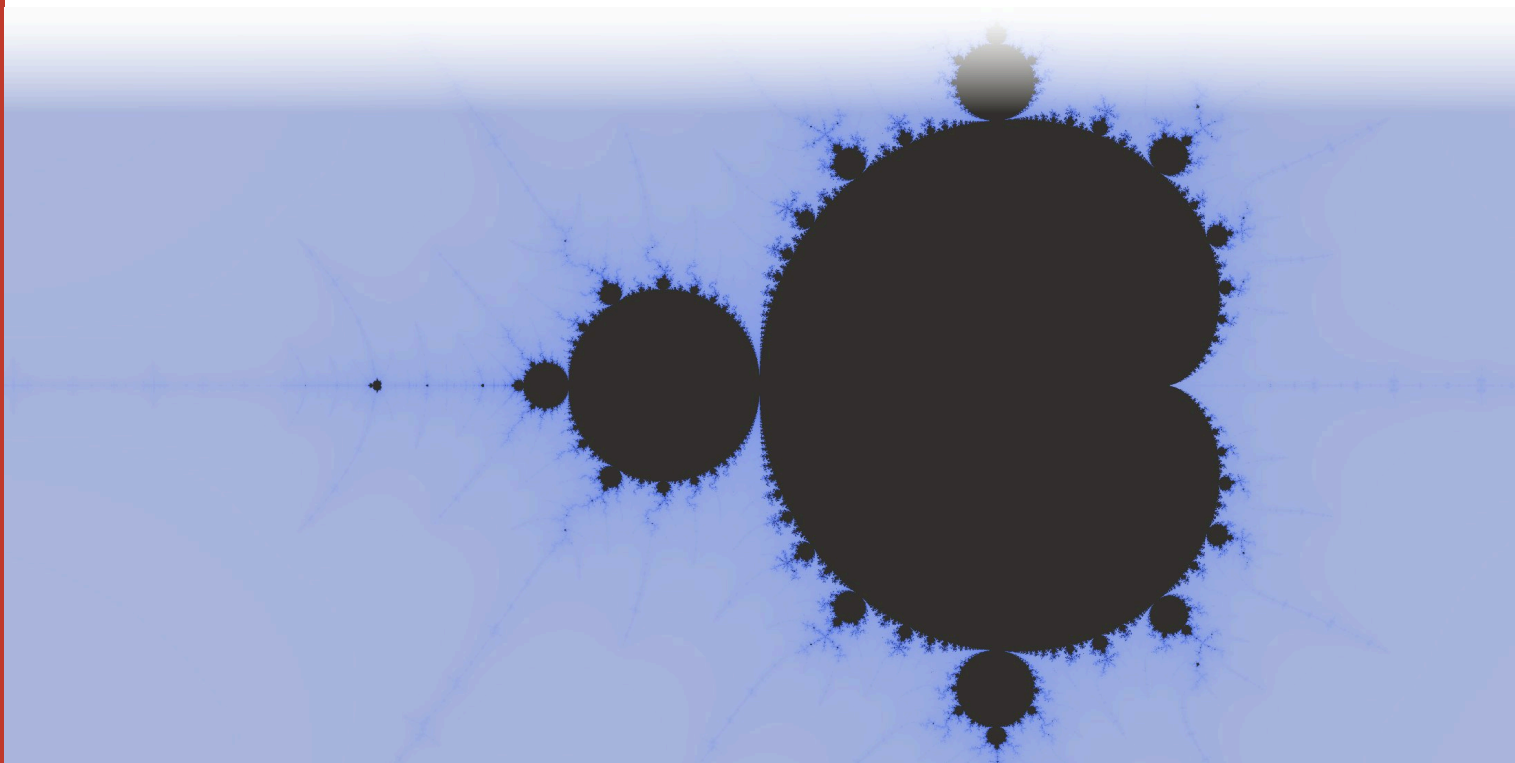
Stage 4 Number and algebra: MA4-INT-C-01 compares, orders and calculates with integers to solve problems.

NAME

CLASS

CHAPTER

1



Homework. Part 1 is **drill practice**: quick questions on the main skill of each lesson in Chapter 1. Write each answer on its line. Part 2 is **mixed revision**, from **Easy** to **Challenging**. Each question is tagged with its lesson, so you can look back at that lesson's summary if you get stuck.

YOU DO On your own: Part 1, drill practice.

QUESTIONS 1–39

1.01 Integers and opposites**Easy** Write the opposite of each integer.

1 12

2 -25

3 -7

4 29

Medium Simplify. The opposite of the opposite is the number itself.

5 $-(-20)$

6 $-(-8)$

7 $-(-60)$

8 $-(-10)$

1.02 The number line**Easy** Which integer is further to the right on the number line?

9 10 or -10

10 -9 or -8

11 1 or -6

12 15 or 13

Medium How many units apart are the two integers on the number line?

13 -13 and 11

14 -3 and -2

15 5 and 6

16 -18 and 11

1.03 Comparing and ordering integers**Easy** Write $<$ or $>$ in the box.

17 $-12 \square 1$

18 $-6 \square -4$

19 $-6 \square 9$

20 $-14 \square -4$

Medium True or false?

21 $4 > -8$

22 $3 < -10$

23 $7 < 9$

24 $-6 > -16$

1.04 Adding integers**Easy** Evaluate each sum.

25 $-8 + 12$

26 $-3 + 8$

27 $-1 + 9$

28 $-10 + 5$

Medium Evaluate. Work from left to right.

29 $6 + (-7) + (-9)$

30 $-8 + 7 + 2$

31 $-12 + 13 + 11$

1.05 Subtracting integers**Easy** Evaluate. Subtract by moving left on the number line.

32 $-9 - 5$

33 $3 - 11$

34 $9 - 13$

35 $-2 - 10$

Medium Evaluate.

36 $-19 - 12$

37 $27 - 36$

38 $-34 - (-2)$

39 $-31 - (-17)$

YOU DO

On your own: Part 1, drill practice.

QUESTIONS 40–71

1.06 Multiplying integers

Easy Evaluate each product. Same signs: positive. Different signs: negative.

(40) $10 \times (-5)$

(41) -6×2

(42) $-6 \times (-5)$

(43) -4×2

Medium Evaluate. Multiply two numbers at a time.

(44) $3 \times (-3) \times (-2)$

(45) $-3 \times (-4) \times 1$

(46) $-3 \times 6 \times 1$

1.07 Dividing integers

Easy Evaluate each quotient. Same signs: positive. Different signs: negative.

(47) $3 \div (-3)$

(48) $24 \div (-3)$

(49) $-70 \div 10$

(50) $36 \div (-9)$

Medium Evaluate. Work from left to right.

(51) $12 \div (-1) \div 3$

(52) $6 \div (-2) \div (-1)$

(53) $32 \div (-4) \div 4$

1.08 Order of operations with integers

Easy Evaluate. $\times$ and $\div$ before $+$ and $-$.

(54) $-7 - (-1) \times 2$

(55) $-9 - 6 \times 4$

(56) $-5 + 5 \times (-5)$

Medium Evaluate. $\times$ and $\div$ from left to right, then $+$ and $-$.

(57) $-3 \times 1 + 9 \div (-3)$

(58) $-2 \times 4 + (-1) \div (-1)$

(59) $5 \times 3 - (-20) \div (-4)$

1.09 Integers on a calculator

Easy Use your calculator. Use the $(-)$ key for negatives.

(60) $-212 + (-56)$

(61) $140 + 81$

(62) $-45 + (-71)$

(63) $-220 + 199$

Medium Use your calculator.

(64) $-378 \div 21$

(65) $-312 \div 39$

(66) $-494 \div 19$

(67) $245 \div (-35)$

1.10 Integer problems

Easy Find the new temperature.

(68) It is -2°C . The temperature rises 6°C .

(69) It is 4°C . The temperature rises 8°C .

Medium Find the difference between the two heights (in metres).

(70) a hill 346 m above sea level and a reef 61 m below

(71) a hill 311 m above sea level and a reef 102 m below

YOU DO On your own: Part 2, mixed revision.

QUESTIONS 72–81

72

Easy

Write the opposite of the integer.

-13 Lesson 1.01

74

Easy

Evaluate each quotient.

$-48 \div 6$ Lesson 1.07

76

Easy

Write $<$ or $>$ in the box.

$-7 \square -4$ Lesson 1.03

78

Easy

Use your calculator.

-64×15 Lesson 1.09

80

Easy

Evaluate.

$-4 - (-9)$ Lesson 1.05

73

Easy

Evaluate each sum.

$-14 + 6$ Lesson 1.04

75

Easy

Heights and depths.

A cave is 35 m below sea level and a hill is 120 m above. Find the difference. Lesson 1.10

77

Easy

Evaluate each product.

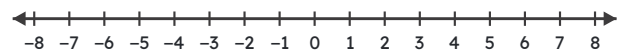
$-11 \times (-3)$ Lesson 1.06

79

Easy

Mark each set of numbers on the number line.

$-7, 4, -4, 0$ Lesson 1.02



81

Easy

Evaluate. $\times$ and $\div$ before $+$ and $-$.

$-5 + 2 \times (-3)$ Lesson 1.08

YOU DO On your own: Part 2, mixed revision.

QUESTIONS 82–91

- 82 **Medium** Write an integer for “a drop of 9°C ”, then write its opposite. Lesson 1.01

- 84 **Medium** $-42 \div (-6)$ Lesson 1.07

- 86 **Medium** Write in descending order: 2, -5, 0, -1 Lesson 1.03

- 88 **Medium** $-15 \times (-7) - 100$ Lesson 1.09

- 90 **Medium** $4 - (-9)$ Lesson 1.05

- 83 **Medium** $-4 + (-8) + 5$ Lesson 1.04

- 85 **Medium** A diver goes from -30 m to -12 m. How far did she rise? Lesson 1.10

- 87 **Medium** $-8 \times (-9)$ Lesson 1.06

- 89 **Medium** Which is further left on a number line: -8 or -2? Lesson 1.02

- 91 **Medium** $-3 \times (4 - 9)$ Lesson 1.08

YOU DO On your own: Part 2, mixed revision.

QUESTIONS 92–96

- 92** **Challenging** The opposite of a number is -31 . What is the number? Explain. Lesson 1.01

- 94** **Challenging** Write in ascending order: -12 , 5 , $-(-8)$, -20 , 0 Lesson 1.03

- 96** **Challenging** The temperature went from -7°C to 12°C . How much did it rise? Lesson 1.05

- 93** **Challenging** $-80 \div (-4) \div (-5)$ Lesson 1.07

- 95** **Challenging** $(-35 - 45) \div (-16)$ Lesson 1.09